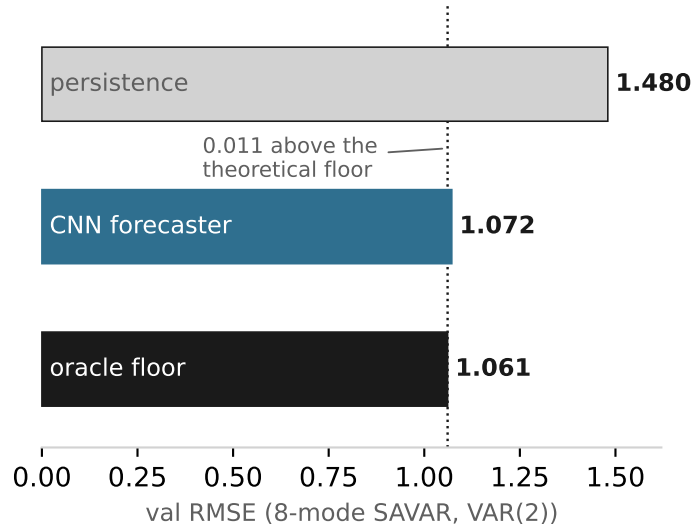
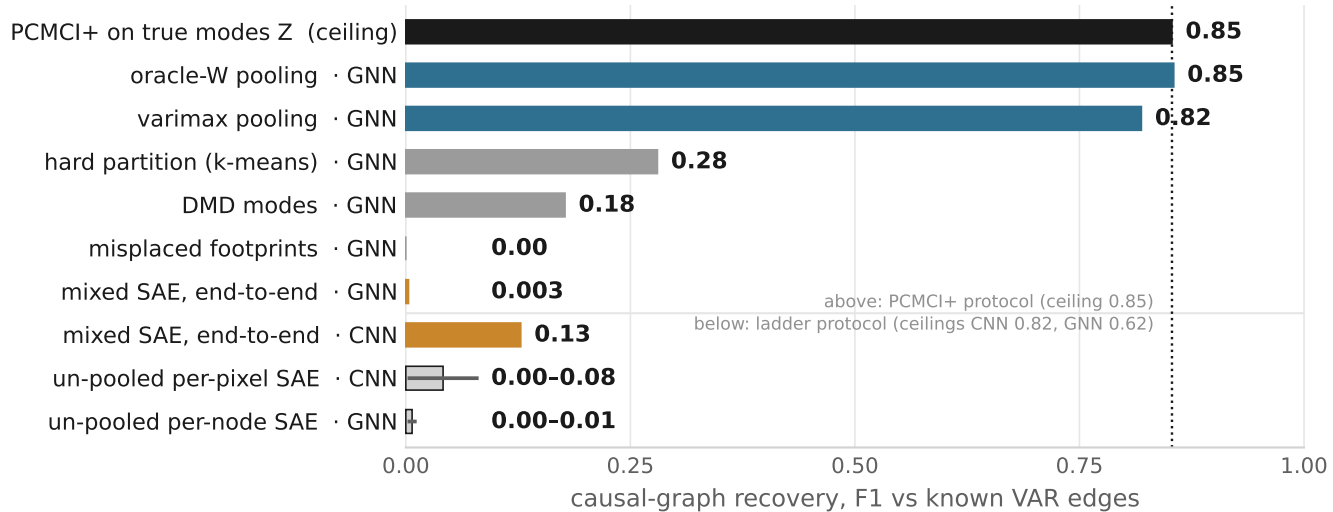


a The forecaster is essentially perfect



b ...but the representation does not expose the causes



Same PCMCi+ estimator throughout; only the variable definition changes. Bake-off rows: GNN forecaster. End-to-end rows: one SAE over mixed activations, N unknown, no footprints — GNN F1 0.003 (0 true edges; $p = 0.31$ vs random feature draws), CNN 0.13 (precision 0.08; $p = 0.03$, single random draws score higher); dropping the spatial pooling too leaves both at 0.00 -0.08 ($p \geq 0.29$).